

mielke-geometrodynamics: residuals

equations measured 2026-09-05T13:17:30Z against the model of 2026-09-01

Corrected (3)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00378 (was)	128	■ 0.136	$D\vartheta^\alpha=R_\beta{}^\alpha\vartheta^\beta=0,\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_\alpha=-\stackrel{(\pm)}{R}_\alpha{}^\beta\wedge\stackrel{(\pm)}{\Pi}_\beta=\pm\frac{i}{2}\ell^2\left(e_\alpha\lrcorner\stackrel{(\pm)}{\Pi}^\beta\right)\wedge\Sigma_\beta\cong0.$		—
mielke-geometrodynamics_E00378 (now)	128	—	preserve the term Pontryagin index for the classification of real associated vector bundles (<code>\textsc{Mayer}</code> & <code>\textsc{Drechsler}</code> 1977). Nevertheless, the denotation of <code>\textsc{Belavin}</code>	<i>preserve the term Pontryagin index for the classification of real associated vector bundles (Mayer & Drechsler 1977). Nevertheless, the denotation of Belavin</i>	—
basis: inferred / ink					
mielke-geometrodynamics_E00402 (was)	134	■ 0.042	$\begin{array}{ll} q: & \underline{*}^\vartheta\Psi_\parallel(\vartheta)=\underline{*}'\vartheta^\beta\Psi_\parallel(\vartheta), \\ \pm p: & \pm\stackrel{*}{\Pi}\Psi_\parallel(\vartheta)=-i\ell^2\frac{\delta}{\delta\stackrel{*}{\alpha}}\Psi_\parallel(\vartheta), \end{array}$		—
mielke-geometrodynamics_E00402 (now)	134	—	$\begin{array}{ll} q: & \stackrel{*}{\vartheta}^\beta\Psi_\parallel(\vartheta)=\stackrel{*'}{\vartheta}^\beta\Psi_\parallel(\vartheta), \\ \pm p: & \stackrel{*}{\Pi}_\alpha\Psi_\parallel(\vartheta)=-i\ell^2\frac{\delta}{\delta\stackrel{*}{\vartheta}_\alpha}\Psi_\parallel(\vartheta), \end{array}$		—
basis: inferred / ink					
mielke-geometrodynamics_E00972 (was)	314	■ 0.153	$\stackrel{(\circ)}{D}\alpha:=\stackrel{(\circ)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\stackrel{(\circ)}{\omega}$		—
mielke-geometrodynamics_E00972 (now)	314	—	$\overset{(1)}{D}\alpha:=\overset{(0)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega^g$		—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_155 E09472 (7.3.11)	155	■ 0.682 ● C +23	\stackrel{+}{\Omega} = 0 \Longleftrightarrow R_{ij} = 0 \Longleftrightarrow R_{ij} = \Lambda' g_{ij}.	(not rendered)	
mielke-geometrodynamics_278 E08857 (12.4.4)	278	■ 0.925 ● C +3	\left.\mathbf{e}_n\right _{\mathscr{G}^A} \cong n (D\mathbf{e}_n),	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_101 E09293 (4.5.4) [conf 0.052]	101	■ 0.052 ● N +0	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D \Psi \right) \wedge \frac{\partial L}{\partial D \Psi}.$	$\xi \downharpoonleft L \equiv (\xi \downharpoonleft \vartheta^\alpha) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + (\xi \downharpoonleft \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downharpoonleft D \Psi) \wedge \frac{\partial L}{\partial D \Psi}.$	
mielke-geometrodynamics_47 E09064 (2.8.6) [conf 0.054]	47	■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	
mielke-geometrodynamics_133 E09400 (6.4.19) [conf 0.060]	133	■ 0.060 ● S +1	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{A}}_\alpha = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{\mathcal{A}}}_\alpha = 0,$	
mielke-geometrodynamics_229 E09718 (10.9.4) [conf 0.066]	229	■ 0.066 ● N +0	$\begin{aligned} & \Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft D \Psi) \wedge \frac{\partial L_\Psi}{\partial D \Psi} \\ & \quad - (e_\alpha \downharpoonleft D \Psi) \wedge \frac{\partial L_\Psi}{\partial D \Psi} - (e_\alpha \downharpoonleft D \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial D \bar{\Psi}}; \end{aligned}$	$\begin{aligned} & \Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial \bar{\Psi}} \\ & \quad - (e_\alpha \downharpoonleft D \Psi) \wedge \frac{\partial L_\Psi}{\partial D \Psi} - (e_\alpha \downharpoonleft D \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial D \bar{\Psi}}; \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_131 E08391 (6.4.6) [conf 0.066]	131	0.066	$\begin{aligned} \Sigma_{\alpha} &:= \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}. \end{aligned}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_168 E08529 (7.7.49) [conf 0.099]	168	0.099	$\begin{aligned} L_{\text{Euler}} &:= \frac{(-1)^{\text{sig}+1}}{2} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right)$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right)$
mielke-geometrodynamics_315 E09974 (14.2.15) [conf 0.099]	315	0.099 N +0	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\pi\overset{\circ}{\Theta} = 0} \right],$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					